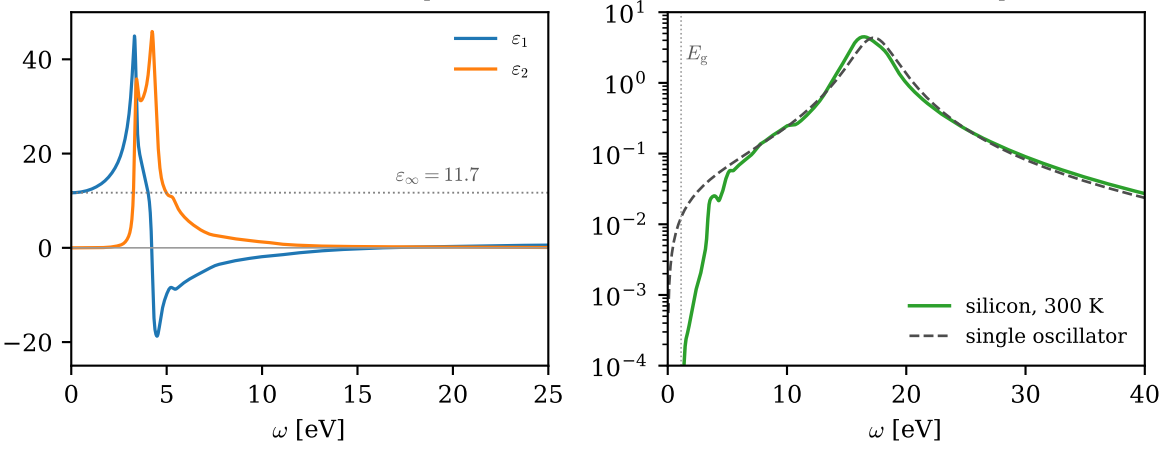
(a) dielectric function, $q \to 0$
ε_1
ε_2
$\varepsilon_\infty = 11.7$
40
20
0
−20
0
5
10
15
20
25
ω [eV]
(b) loss function L, $q \to 0$
E_g
10^1
10^0
10^{-1}
10^{-2}
10^{-3}
10^{-4}
silicon, 300 K
single oscillator
0
10
20
30
40
ω [eV]